

Lecture 3: Asymptotics of Random Experiments and Concentration Inequalities

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1 The Bridge between Probability and Statistics: Laws of Large Numbers and Central Limit Theorem

So far, our study has focused on the theoretical properties of probability distributions. We have assumed knowledge of the underlying parameters (like p in a Bernoulli trial or μ and σ^2 in a Normal distribution) and deduced the probabilities of certain outcomes.

In the real world, the situation is reversed. We do not know the true parameters. We cannot observe the entire population, only a **sample**. We might observe a sequence of 1000 coin flips, but we do not know the true bias p of the coin. We might measure the heights of 500 people, but we do not know the true average height μ of the entire population.

Statistics addresses this inverse problem: how can we infer the properties of an underlying probability distribution from a finite sample of data?

This lecture builds the theoretical bridge that justifies this inference. We will explore two of the most important theorems in all of probability theory:

1. **The Law of Large Numbers (LLN):** This theorem tells us that the average of a large number of independent trials will converge to its theoretical expected value. It is the reason we can be confident that the average height of our sample is a good estimate of the true average height of the population.
2. **The Central Limit Theorem (CLT):** This theorem goes a step further. It describes the *distribution* of the error between the sample average and the true expected value. It tells us that this error, when properly scaled, follows a Normal distribution, regardless of the original distribution of the data.

1.1 The Sample Mean and its Properties

The central object of study in this section is the average of our observations.

Definition 1 (Sample Mean). Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed (i.i.d.) random variables, each with mean $\mu = \mathbb{E}[X_i]$ and variance $\sigma^2 = \text{Var}(X_i)$. The **sample mean** (or empirical average) is defined as:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

Before we can prove that $\bar{X}_n$ converges to μ , we must first understand its own mean and variance.

Proposition 1 (Mean and Variance of the Sample Mean). The expected value of the sample mean is the true mean μ , and its variance is σ^2/n .

1. $\mathbb{E}[\bar{X}_n] = \mu$
2. $\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$

Proof. 1. Using the linearity of expectation:

$$\mathbb{E}[\bar{X}_n] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] = \frac{1}{n} \sum_{i=1}^n \mu = \frac{1}{n} (n\mu) = \mu.$$

This shows that the sample mean is an **unbiased estimator** of the true mean.

2. Using the properties of variance for independent random variables:

$$\text{Var}(\bar{X}_n) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n^2} \text{Var}\left(\sum_{i=1}^n X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \sum_{i=1}^n \sigma^2 = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$$

This second result is crucial: as the sample size n increases, the variance of the sample mean decreases. This means that $\bar{X}_n$ becomes more and more “concentrated” around its mean μ . The laws of large numbers make this notion precise. $\square$

To prove the LLN, we need tools that can bound the probability of a random variable deviating from its mean.

1.2 Chebyshev's Inequality

Chebyshev's inequality provides a bound using the variance. It is a direct consequence of Markov's inequality.

Theorem 2 (Chebyshev's Inequality). *Let Y be a random variable with finite mean μ_Y and finite variance σ_Y^2 . Then for any constant $k > 0$:*

$$\mathbb{P}(|Y - \mu_Y| \geq k) \leq \frac{\sigma_Y^2}{k^2}.$$

Proof. Consider the non-negative random variable $Z := (Y - \mu_Y)^2$. Its expectation is $\mathbb{E}[Z] = \mathbb{E}[(Y - \mu_Y)^2] = \sigma_Y^2$. The event $|Y - \mu_Y| \geq k$ is equivalent to the event $(Y - \mu_Y)^2 \geq k^2$, which is $Z \geq k^2$. We can now apply Markov's inequality to Z with $a = k^2$:

$$\mathbb{P}(Z \geq k^2) \leq \frac{\mathbb{E}[Z]}{k^2}.$$

Substituting back, we get:

$$\mathbb{P}(|Y - \mu_Y| \geq k) \leq \frac{\sigma_Y^2}{k^2}.$$

□

1.3 The laws of large numbers

The weak law of large numbers states that as the sample size increases, the probability that the sample mean is further than some small amount ϵ from the true mean goes to zero. This is called **convergence in probability**.

Theorem 3 (The Weak Law of Large Numbers (WLLN)). *Let $X_1, \dots, X_n$ be i.i.d. random variables with mean μ and finite variance σ^2 . Then for any $\epsilon > 0$:*

$$\lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) = 0$$

We denote this as $\bar{X}_n \xrightarrow{p} \mu$.

Proof. The proof is a direct and elegant application of Chebyshev's inequality to the sample mean $\bar{X}_n$. Let $Y = \bar{X}_n$. We know its mean is $\mu_Y = \mathbb{E}[\bar{X}_n] = \mu$ and its variance is $\sigma_Y^2 = \text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$. Applying Chebyshev's inequality with $k = \epsilon$:

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\epsilon^2} = \frac{\sigma^2/n}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2}.$$

As we take the limit as $n \rightarrow \infty$:

$$\lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X}_n - \mu| \geq \epsilon) \leq \lim_{n \rightarrow \infty} \frac{\sigma^2}{n\epsilon^2} = 0$$

Since probability cannot be negative, the limit must be exactly 0. □

The weak law of large numbers states that for any large n , a significant deviation from the mean is *unlikely*, but it does not rule out the possibility that such deviations, however unlikely, could happen infinitely often. The strong law of large numbers rules this out. It says the sequence of outcomes $\bar{X}_n(\omega)$ will actually converge to μ for all outcomes ω in a set of probability 1. This is called **almost sure convergence**.

Theorem 4 (Kolmogorov's Strong Law of Large Numbers (SLLN)). *Let $X_1, \dots, X_n$ be i.i.d. random variables. $\bar{X}_n$ converges to μ almost surely if and only if $\mathbb{E}[|X_1|] =: \mu$ is finite:*

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1$$

We denote this as $\bar{X}_n \xrightarrow{a.s.} \mu$.

The Monte Carlo Method for Estimating π : A powerful and concrete application of the Laws of Large Numbers is the family of **Monte Carlo methods**, which use repeated random sampling to obtain numerical results. The WLLN provides the theoretical guarantee that these methods work. A classic illustration is the estimation of the value of π .

Consider a square in the 2D Cartesian plane with vertices at $(-1, 1)$, $(1, 1)$, $(1, -1)$, and $(-1, -1)$. This square has a side length of 2 and an area of $A_{\text{square}} = 4$. Inside this square, we inscribe a circle of radius $r = 1$, centered at the origin. The area of this circle is $A_{\text{circle}} = \pi r^2 = \pi$.

If we throw a “dart” that lands at a random, uniformly distributed point within the square, the probability that it also lands inside the circle is the ratio of the areas:

$$p = \frac{A_{\text{circle}}}{A_{\text{square}}} = \frac{\pi}{4}.$$

We can model this experiment using random variables.

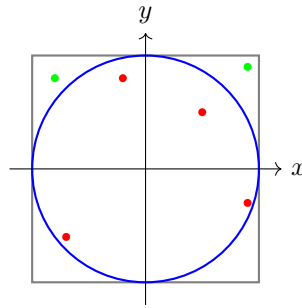


Figure 1: A square of area 4 with an inscribed circle of area π . Points are sampled uniformly within the square.

- Let $X \sim U[-1, 1]$ and $Y \sim U[-1, 1]$ be two independent random variables representing the coordinates of a random point.
- We define a Bernoulli random variable, Z , for each point we generate:

$$Z = \begin{cases} 1 & \text{if } X^2 + Y^2 \leq 1 \quad (\text{the point is in the circle}) \\ 0 & \text{if } X^2 + Y^2 > 1 \quad (\text{the point is outside the circle}) \end{cases}$$

- The expected value of this Bernoulli random variable is precisely the probability of success, p :

$$\mathbb{E}[Z] = 1 \cdot \mathbb{P}(Z = 1) + 0 \cdot \mathbb{P}(Z = 0) = \mathbb{P}(X^2 + Y^2 \leq 1) = p = \frac{\pi}{4}.$$

The true mean, μ , of our random variable Z is the unknown quantity $\pi/4$.

Now, we perform this experiment n times, generating a sequence of independent and identically distributed (i.i.d.) random variables $Z_1, Z_2, \dots, Z_n$. We then calculate their sample mean:

$$\bar{Z}_n = \frac{1}{n} \sum_{i=1}^n Z_i.$$

This sample mean is simply the **proportion of darts that landed inside the circle**. The Weak Law of Large Numbers states that this sample mean converges in probability to the true mean:

$$\bar{Z}_n \xrightarrow{p} \mathbb{E}[Z]$$

For any small error tolerance $\epsilon > 0$, this means:

$$\lim_{n \rightarrow \infty} P\left(\left|\bar{Z}_n - \frac{\pi}{4}\right| \geq \epsilon\right) = 0.$$

Conclusion: The WLLN provides the theoretical justification that for a large number of trials n , our empirical proportion of “hits”, $\bar{Z}_n$, is very likely to be very close to the true probability, $\pi/4$. We can therefore use our simulation to construct an estimator for π :

$$4 \cdot \bar{Z}_n = 4 \cdot \frac{\text{Number of Hits inside Circle}}{\text{Total Number of Trials}} \xrightarrow{p} \pi.$$

This is a direct and concrete application: the WLLN guarantees that a simple, repeated random experiment can be used to reliably estimate a complex analytical quantity. This is the foundational principle behind all Monte Carlo simulations.

1.4 The Central Limit Theorem

The laws of large numbers are powerful results that guarantee the convergence of the sample mean to the true mean. They tell us *that* the sample average gets closer to the target:

$$\bar{X}_n \xrightarrow{p} \mu$$

However, they do not tell us anything about the *manner* of this convergence. For a large but finite sample size n , what is the distribution of the error term, $\bar{X}_n - \mu$? How likely are we to be a certain distance away from the mean?

The Central Limit Theorem (CLT) answers this question. It is perhaps the most remarkable result in all of probability theory. It states that the distribution of the sample mean, when properly centered and scaled, converges to a standard Normal (Gaussian) distribution, **regardless of the original distribution of the data**.

An intuitive analogy is that of an archer:

- **The LLN** states that as the archer shoots more arrows, the average position of all their arrows will get closer and closer to the bullseye.
- **The CLT** states that if you look at the distribution of the arrow holes on the target, they will form a bell-shaped pattern around the bullseye, even if the archer's individual actions (muscle twitches, breathing, etc.) follow very non-normal distributions.

The error $\bar{X}_n - \mu$ shrinks as n grows. To study its distribution, we must “zoom in” by scaling it. The correct scaling factor is $\sqrt{n}$, which counteracts the shrinking standard deviation of the sample mean ($\sigma/\sqrt{n}$).

1.5 Formal Statement of the Theorem

Theorem 5 (The Lindeberg-Lévy Central Limit Theorem). *Let $X_1, X_2, \dots, X_n$ be a sequence of independent and identically distributed (i.i.d.) random variables, each with a finite mean μ and a finite, non-zero variance σ^2 .*

Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ be the sample mean. Then the standardized sample mean

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}$$

converges in distribution to a standard normal random variable $Z \sim N(0, 1)$.

$$Z_n \xrightarrow{d} Z.$$

Convergence in distribution means that the Cumulative Distribution Function (CDF) of Z_n converges to the CDF of the standard normal distribution $N(0, 1)$, denoted $\Phi(z)$. For every $z \in \mathbb{R}$:

$$\lim_{n \rightarrow \infty} \mathbb{P}(Z_n \leq z) = \Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

In practice, this means that for a sufficiently large n (often a rule of thumb is $n \geq 30$), we can use the Normal distribution as an excellent approximation for the distribution of the sample mean:

$$\bar{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right).$$

Statistical Quality Control: The true power of the CLT is that it allows us to compute probabilities for sample means even when we have no knowledge of the underlying distribution of the individual data points.

A factory produces lightbulbs, and the lifetime of any given lightbulb is a random variable. The manufacturing process has been stable for years, and historical data shows that the true mean lifetime is $\mu = 800$ hours with a true standard deviation of $\sigma = 100$ hours. The actual distribution of lifetimes is unknown and may be skewed (e.g., most bulbs last a long time, but some fail very early).

The quality control department pulls a random sample of $n = 100$ lightbulbs for testing. What is the probability that the **average lifetime** of this sample is less than 775 hours?

Approach:

We want to calculate $\mathbb{P}(\bar{X}_{100} < 775)$. We do not know the distribution of the individual lifetimes X_i , so we cannot calculate this probability directly.

Since the sample size $n = 100$ is large, the CLT tells us that the distribution of the sample mean $\bar{X}_{100}$ can be approximated by a Normal distribution.

- The mean of this distribution is $\mathbb{E}[\bar{X}_{100}] = \mu = 800$.
- The standard deviation of this distribution (called the standard error) is $\text{SD}(\bar{X}_{100}) = \frac{\sigma}{\sqrt{n}} = \frac{100}{\sqrt{100}} = 10$.

So, we can model the sample mean as:

$$\bar{X}_{100} \approx N(\text{mean} = 800, \text{std dev} = 10)$$

To find the probability, we convert our value of 775 to a standard Z-score:

$$Z = \frac{\bar{X}_{100} - \mu}{\sigma/\sqrt{n}} < \frac{775 - 800}{10} = \frac{-25}{10} = -2.5.$$

Therefore, the probability we want to find is equivalent to finding the probability that a standard normal random variable is less than -2.5:

$$\mathbb{P}(\bar{X}_{100} < 775) \approx \mathbb{P}(Z < -2.5),$$

where $Z \sim N(0, 1)$. Using a standard normal table or calculator, we find the value of the CDF $\Phi(-2.5)$:

$$\Phi(-2.5) \approx 0.0062.$$

Conclusion:

There is only about a 0.62% chance that the average lifetime of a sample of 100 bulbs will be 775 hours or less. If the quality control department observes such a result, they would have strong evidence to suspect that the manufacturing process has changed for the worse (i.e., that the true mean μ is no longer 800).

This example is powerful because we reached a very specific, quantitative conclusion **without ever knowing the shape of the original distribution** of lightbulb lifetimes.

2 Concentration Inequalities

The Law of Large Numbers and the Central Limit Theorem are **asymptotic** results. They describe the behavior of the sample mean as the sample size n approaches infinity. In practice, however, we always work with a finite sample size n .

This raises a crucial question: for a given finite n , how “concentrated” is the sample mean around its expected value? How can we bound the probability of observing a large deviation from the mean?

Concentration inequalities provide these non-asymptotic bounds. They are the theoretical foundation for many results in statistical learning theory, randomized algorithms, and high-dimensional statistics. While Chebyshev’s inequality gives a polynomial bound on this deviation probability ($\mathcal{O}(1/n)$), the inequalities in this section provide much stronger **exponential bounds** ($\mathcal{O}(e^{-cn})$).

2.1 Moment-Generating Functions (MGFs)

We start by defining a key tool: the Moment-Generating Function (MGF). The MGF is an alternative way to specify a probability distribution that is particularly useful for studying sums of independent random variables.

Definition 2 (Moment-Generating Function). *The **moment-generating function (MGF)** of a random variable X , denoted $M_X(t)$, is defined as:*

$$M_X(t) = \mathbb{E}[e^{tX}]$$

for all real numbers t for which this expectation exists.

The MGF is named for its ability to “generate” the moments of the distribution. The k -th moment, $\mathbb{E}[X^k]$, can be found by taking the k -th derivative of the MGF with respect to t and evaluating it at $t = 0$:

$$\mathbb{E}[X^k] = \left. \frac{d^k}{dt^k} M_X(t) \right|_{t=0}.$$

For our purposes in concentration inequalities, the most important property of the MGF is how it behaves with sums of independent random variables. If $S_n = \sum_{i=1}^n X_i$ is a sum of independent variables, then the MGF of the sum is the product of the individual MGFs:

$$M_{S_n}(t) = \mathbb{E}[e^{tS_n}] = \prod_{i=1}^n M_{X_i}(t).$$

This property, which turns a sum into a product, is what makes the MGF the perfect tool for studying the, so-called, Chernoff bound.

2.2 The Chernoff Bound: The General Method

The Chernoff bound is not a single inequality, but rather a general *method* for deriving tight bounds on the tail probabilities of a sum of independent random variables. The method relies on a clever application of Markov’s inequality to the moment-generating function.

Theorem 6 (The Chernoff Bound Method). *Let $S_n = \sum_{i=1}^n X_i$ be a sum of independent (but not necessarily identically distributed) random variables. For any constant a :*

$$\mathbb{P}(S_n \geq a) \leq \inf_{t>0} e^{-ta} \mathbb{E}[e^{tS_n}].$$

If the X_i are i.i.d., this simplifies to:

$$\mathbb{P}(S_n \geq a) \leq \inf_{t>0} e^{-ta} (M_X(t))^n$$

where $M_X(t)$ is the MGF of a single random variable X_i .

Proof. The proof is known as the “Chernoff trick”.

1. For any $t > 0$, the function $x \mapsto e^{tx}$ is monotonically increasing. Therefore, the event $S_n \geq a$ is identical to the event $e^{tS_n} \geq e^{ta}$.

$$\mathbb{P}(S_n \geq a) = \mathbb{P}(e^{tS_n} \geq e^{ta}).$$

2. Let $Y := e^{tS_n}$. Y is a non-negative random variable, thus we can apply Markov’s inequality:

$$\mathbb{P}(Y \geq e^{ta}) \leq \frac{\mathbb{E}[Y]}{e^{ta}}.$$

Substituting back gives:

$$\mathbb{P}(S_n \geq a) \leq \frac{\mathbb{E}[e^{tS_n}]}{e^{ta}} = e^{-ta} \mathbb{E}[e^{tS_n}].$$

3. This inequality holds for *any* choice of $t > 0$. To get the tightest possible bound, we are free to choose the value of t that minimizes the right-hand side. This is the infimum:

$$\mathbb{P}(S_n \geq a) \leq \inf_{t>0} e^{-ta} \mathbb{E}[e^{tS_n}].$$

By independence of the X_i , we have $\mathbb{E}[e^{tS_n}] = \mathbb{E}[e^{t \sum X_i}] = \mathbb{E}[\prod e^{tX_i}] = \prod \mathbb{E}[e^{tX_i}]$. If the variables X_i are also identically distributed, this becomes $(\mathbb{E}[e^{tX_1}])^n = (M_X(t))^n$. $\square$

The Chernoff method is the template from which more specific and user-friendly inequalities, like Hoeffding's, are derived.

2.3 Hoeffding's Inequality: Bounded Random Variables

Hoeffding's inequality is arguably the most widely used concentration inequality in data science and machine learning. It applies the Chernoff method to the important case of sums of independent, **bounded** random variables.

To prove it, we first need a key lemma.

Lemma 7 (Hoeffding's Lemma). *Let X be a random variable with $\mathbb{E}[X] = 0$ and such that $X \in [a, b]$ with probability 1. Then for any $s \in \mathbb{R}$:*

$$\mathbb{E}[e^{sX}] \leq e^{s^2(b-a)^2/8}.$$

Proof. (Sketch) The proof uses the convexity of the exponential function. Since $X \in [a, b]$, we can write it as a convex combination of a and b . By convexity, $e^{sX} \leq \frac{X-a}{b-a} e^{sb} + \frac{b-X}{b-a} e^{sa}$. Taking the expectation of both sides (and using $\mathbb{E}[X] = 0$) and performing some algebraic manipulation leads to the desired bound. $\square$

Theorem 8 (Hoeffding's Inequality). *Let $X_1, \dots, X_n$ be independent random variables such that $X_i \in [a_i, b_i]$ with probability 1. Let $S_n = \sum_{i=1}^n X_i$. Then for any $\epsilon > 0$:*

$$\mathbb{P}(|S_n - \mathbb{E}[S_n]| \geq \epsilon) \leq 2 \exp \left(\frac{-2\epsilon^2}{\sum_{i=1}^n (b_i - a_i)^2} \right)$$

Proof. We follow the Chernoff recipe. Let $Y_i = X_i - \mathbb{E}[X_i]$ be the centered variables. Then $\mathbb{E}[Y_i] = 0$ and $Y_i \in [a_i - \mathbb{E}[X_i], b_i - \mathbb{E}[X_i]]$, an interval of length $b_i - a_i$. We want to bound $\mathbb{P}(\sum Y_i \geq \epsilon)$.

1. For any $t > 0$, $\mathbb{P}(\sum Y_i \geq \epsilon) \leq e^{-t\epsilon} \mathbb{E}[e^{t \sum Y_i}] = e^{-t\epsilon} \prod_{i=1}^n \mathbb{E}[e^{tY_i}]$.
2. Apply Hoeffding's Lemma to each $\mathbb{E}[e^{tY_i}]$ term:

$$\mathbb{E}[e^{tY_i}] \leq e^{t^2(b_i - a_i)^2/8}.$$

3. Substitute this back into the bound:

$$\mathbb{P} \left(\sum Y_i \geq \epsilon \right) \leq e^{-t\epsilon} \prod_{i=1}^n e^{t^2(b_i - a_i)^2/8} = \exp \left(-t\epsilon + \frac{t^2}{8} \sum_{i=1}^n (b_i - a_i)^2 \right).$$

4. Minimize the exponent with respect to t . The minimum of $-t\epsilon + Ct^2$ occurs at $t = \epsilon/(2C)$, where $C := \frac{1}{8} \sum (b_i - a_i)^2$. So the optimal $t = \frac{4\epsilon}{\sum (b_i - a_i)^2}$.
5. Plugging this optimal t back in yields the one-sided bound:

$$\mathbb{P}(S_n - \mathbb{E}[S_n] \geq \epsilon) \leq \exp \left(\frac{-2\epsilon^2}{\sum (b_i - a_i)^2} \right)$$

6. The two-sided bound is obtained by applying the same logic to $\mathbb{P}(-\sum Y_i \geq \epsilon)$ and adding the two probabilities using a union bound, which multiplies the result by 2.

$\square$

Generalization Error in Machine Learning: Hoeffding's inequality is the foundation for proving that a model that performs well on a test set will likely perform well on future, unseen data.

- Let's say we have a binary classifier (e.g., spam vs. not spam). Let E_{true} be the true, unknown error rate of our classifier on all possible data.
- We test our model on a test set of n i.i.d. data points. Let E_{test} be the observed error rate on this set.
- For each test point i , we can define a Bernoulli random variable X_i : $X_i = 1$ if the classifier is wrong, $X_i = 0$ if it's right.
- The variables X_i are bounded in $[0, 1]$. The true error is $\mathbb{E}[X_i] = E_{true}$. The test error is the sample mean, $E_{test} = \bar{X}_n$.
- Hoeffding's inequality for the sample mean states:

$$\mathbb{P}(|E_{test} - E_{true}| \geq \epsilon) \leq 2 \exp(-2n\epsilon^2).$$

This provides a powerful guarantee: the probability that our measured test error is far from the true error decreases exponentially with the size of the test set. This allows us to be confident in our model evaluation. For example, to be 95% sure that our measured error is within $\epsilon = 0.01$ of the true error, we can solve $2e^{-2n(0.01)^2} \leq 0.05$ for n .

2.4 Bernstein's Inequality: Using Variance Information

Hoeffding's inequality is powerful but can be loose if the variance of the random variables is small compared to their range. Bernstein's inequality is a refinement that incorporates variance information to provide a tighter bound in such cases.

Theorem 9 (Bernstein's Inequality). *Let $X_1, \dots, X_n$ be independent random variables with $\mathbb{E}[X_i] = 0$. Suppose that for all i , $|X_i| \leq M$ (they are bounded). Let $\sigma_i^2 = \text{Var}(X_i)$ and let $V = \sum_{i=1}^n \sigma_i^2$. Then for any $\epsilon > 0$:*

$$\mathbb{P}\left(\sum_{i=1}^n X_i \geq \epsilon\right) \leq \exp\left(-\frac{\epsilon^2/2}{V + M\epsilon/3}\right)$$

Proof. The proof follows the same Chernoff recipe, but uses a more sophisticated bound on the MGF of a single X_i that takes both its variance and its bound M into account. The full proof is more involved and is omitted here, but the key is that it smoothly interpolates between a sub-Gaussian behavior (when variance dominates) and a sub-Poisson behavior (when the bound M dominates). $\square$

Anomaly detection: Let's quantify the difference between the Hoeffding and Bernstein inequalities in a concrete scenario. A server farm processes millions of transactions per day. The vast majority are normal, but a tiny fraction might be fraudulent.

Scenario:

- Let X_i be a Bernoulli variable for the i -th transaction: $X_i = 1$ if fraudulent, $X_i = 0$ otherwise.
- The true fraud rate is very low: $p = \mathbb{P}(X_i = 1) = 0.001$.
- We observe $n = 1,000,000$ transactions in a day.
- The expected number of fraudulent transactions is $\mathbb{E}[S_n] = np = 1000$.
- **Question:** What is the probability of observing 2000 or more fraudulent transactions, i.e., a deviation of $\epsilon = 1000$ from the mean? We want to bound $\mathbb{P}(S_n - \mathbb{E}[S_n] \geq 1000)$.

Applying Hoeffding's Inequality: The variables X_i are in $[0, 1]$, so the range is $b_i - a_i = 1$. We use the one-sided version from the proof:

$$\mathbb{P}(S_n - \mathbb{E}[S_n] \geq \epsilon) \leq \exp\left(\frac{-2\epsilon^2}{\sum_{i=1}^n (b_i - a_i)^2}\right) = \exp\left(\frac{-2\epsilon^2}{n}\right).$$

Plugging in the numbers:

$$\mathbb{P}(S_n - \mathbb{E}[S_n] \geq 1000) \leq \exp\left(\frac{-2 \cdot (1000)^2}{1,000,000}\right) = \exp(-2) \approx 0.1353.$$

Hoeffding's inequality tells us there is up to a 13.5% chance of this happening. This is a non-trivial probability.

Applying Bernstein's Inequality: We use the centered variables $Y_i = X_i - p$.

- The variables are bounded by $M = \max(p, 1 - p) = 0.999 \leq 1$.

- The total variance is $V = np(1-p) = 1,000,000 \cdot 0.001 \cdot 0.999 = 999$.
- The deviation is $\epsilon = 1000$.

The bound is:

$$\mathbb{P}(S_n - \mathbb{E}[S_n] \geq \epsilon) \leq \exp\left(-\frac{\epsilon^2/2}{V + M\epsilon/3}\right)$$

Plugging in the numbers:

$$\begin{aligned} \mathbb{P}(S_n - \mathbb{E}[S_n] \geq 1000) &\leq \exp\left(-\frac{(1000)^2/2}{999 + (1 \cdot 1000/3)}\right) \\ &\leq \exp\left(-\frac{500,000}{999 + 333.33}\right) \\ &\leq \exp\left(-\frac{500,000}{1332.33}\right) \approx \exp(-375.28) \end{aligned}$$

This value is astronomically small, effectively zero.

Conclusion:

- **Hoeffding:** Provides a loose bound of 13.5%. It is “unaware” of the low event probability, only considering the worst-case range of $[0, 1]$.
- **Bernstein:** Provides an extremely tight bound of effectively zero. By incorporating the small total variance ($V = 999$), it correctly captures our intuition that observing double the expected number of very rare events is almost impossible.

In data science applications involving rare events, like fraud detection, anomaly detection, or estimating CTR for low-performing ads, Bernstein’s inequality provides a much more realistic and useful performance guarantee.

Sample covariance matrix: Let $X = (X_1, \dots, X_d)^T$ be a d -dimensional random vector drawn from a multivariate Gaussian distribution with a zero mean and a true covariance matrix Σ . We denote this as:

$$X \sim N(0, \Sigma).$$

The covariance matrix $\Sigma \in \mathbb{R}^{d \times d}$ is a symmetric positive semidefinite matrix where $\Sigma_{ij} = \text{Cov}(X_i, X_j) = \mathbb{E}[X_i X_j]$. The assumption of a zero mean is made for simplicity and is standard in this analysis.

We are given n independent and identically distributed (i.i.d.) samples, $X^{(1)}, X^{(2)}, \dots, X^{(n)}$, each drawn from this distribution. Our goal is to estimate Σ using the **sample covariance matrix**, $\hat{\Sigma}$:

$$\hat{\Sigma} = \frac{1}{n} \sum_{k=1}^n X^{(k)} (X^{(k)})^T.$$

We want to quantify how close $\hat{\Sigma}$ is to Σ . We will measure this difference using the matrix operator norm, denoted $\|\cdot\|_2$, which corresponds to the largest singular value of a matrix.

To apply concentration inequalities, we define a sequence of random matrices $Z_1, \dots, Z_n$ where each Z_k is formed from a single data sample:

$$Z_k = X^{(k)} (X^{(k)})^T.$$

By construction, our estimator $\hat{\Sigma}$ is the average of these matrices: $\hat{\Sigma} = \frac{1}{n} \sum_{k=1}^n Z_k$.

The key properties of this sequence are:

1. **Independence:** The Z_k matrices are i.i.d. because the $X^{(k)}$ vectors are i.i.d.
2. **Expectation:** The expectation of each random matrix is the true covariance matrix:

$$\mathbb{E}[Z_k] = \mathbb{E}[X^{(k)} (X^{(k)})^T] = \Sigma.$$

The problem is now to bound the deviation of the sample mean of i.i.d. random matrices from their true mean, i.e., to bound $\|\hat{\Sigma} - \Sigma\|_2$. For this, we can use a specialized version of the Matrix Bernstein inequality:

Theorem 10 (Concentration of the Sample Covariance Matrix). *Let $\hat{\Sigma} = \frac{1}{n} \sum_{k=1}^n X^{(k)} (X^{(k)})^T$ be the sample covariance matrix from n i.i.d. samples of $X \sim N(0, \Sigma)$. Then there exists a universal constant $c > 0$ such that for any $\epsilon \in (0, 1)$,*

$$\mathbb{P}\left(\|\hat{\Sigma} - \Sigma\|_2 \geq \epsilon \|\Sigma\|_2\right) \leq 2 \exp(-c n \epsilon^2).$$

Remark 1. This is a powerful **relative error bound**. It states that the error $\|\hat{\Sigma} - \Sigma\|_2$ is small relative to the norm of the true matrix $\|\Sigma\|_2$. Notice that the dimension d does not appear in the exponential term, although it is implicitly present in the operator norms. More advanced results (e.g., from Joel Tropp) provide bounds that explicitly depend on the dimension and can be sharper in some regimes.

The theorem provides a direct answer to the question: “How many samples do I need for a good estimate?” We can rephrase the result to find the required sample size n .

Suppose we want the estimation error $\|\hat{\Sigma} - \Sigma\|_2$ to be at most $\epsilon\|\Sigma\|_2$ with a high probability, say at least $1 - \delta$. We set the failure probability to be at most δ :

$$2 \exp(-cn\epsilon^2) \leq \delta$$

Now, we solve for n :

$$\begin{aligned} \exp(-cn\epsilon^2) &\leq \frac{\delta}{2} \\ -cn\epsilon^2 &\leq \log\left(\frac{\delta}{2}\right) \\ -cn\epsilon^2 &\leq -\log\left(\frac{2}{\delta}\right) \\ n &\geq \frac{1}{c\epsilon^2} \log\left(\frac{2}{\delta}\right). \end{aligned}$$

Corollary 11 (Sample Complexity). To guarantee that $\|\hat{\Sigma} - \Sigma\|_2 \leq \epsilon\|\Sigma\|_2$ with probability at least $1 - \delta$, it suffices to take a number of samples n such that:

$$n \geq C \cdot \frac{\log(2/\delta)}{\epsilon^2}$$

where $C = 1/c$ is a universal constant.